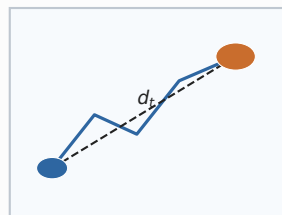


● robot / microrobot ● patient hand

a

Simulation environment



Observation
 O_t

ZPD band
boundaries
smoothness
Reward
 r_t

observation O_t , reward r_t

action a_t

b

Policy learning

adaptive
robot policy



virtual patient pool

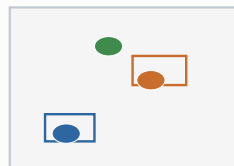
PFSP
sampling

PPO update

control command

c

Vision and state estimation



camera frame

YOLO +
MediaPipe

state
estimator

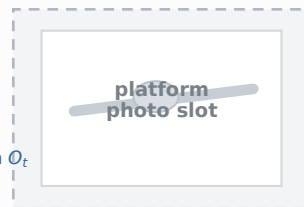
state = positions + velocity
with dead-reckoning gaps

image feed

real-world observation O_t

d

Real-world rehabilitation platform



insert annotated real setup photo

UR10
controller

magnetic
actuation